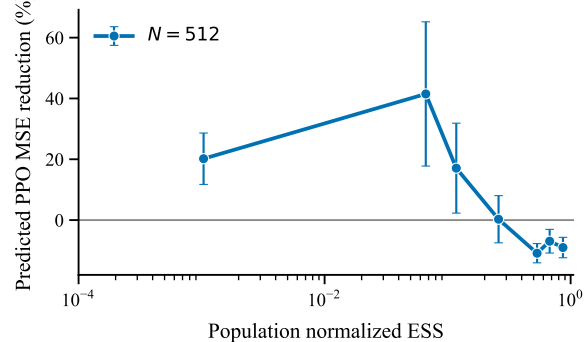
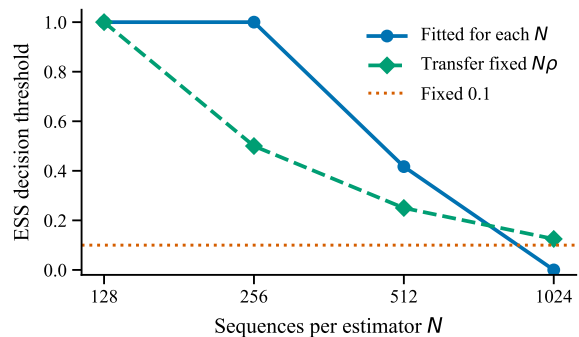
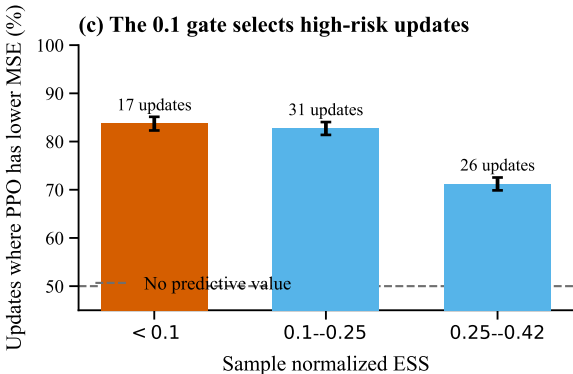


(a) Predicted crossover at informative N **(b) The boundary moves with N** **(c) The 0.1 gate selects high-risk updates****(d) Four-epoch optimization at $N = 512$** 